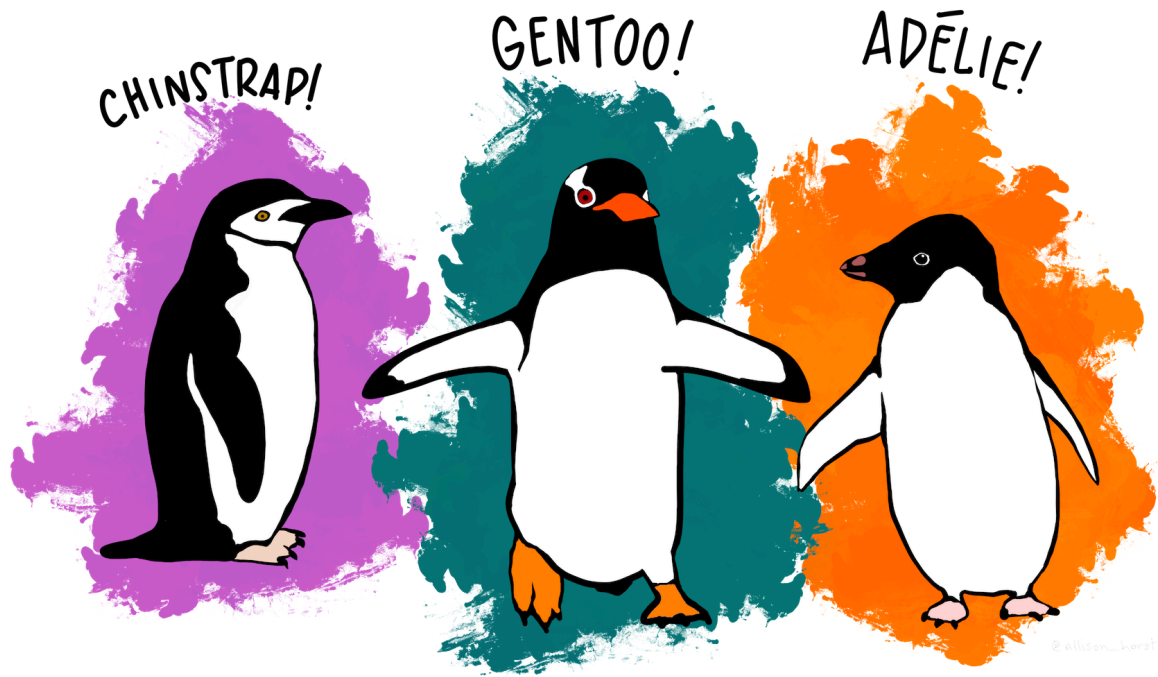


STA 160 Mid-Quarter Project

Palmer Penguin Niche Partitioning: A Data-Driven Analysis of Morphological Differentiation



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Abstract

The Palmer Archipelago has three kinds of penguin: Adelie (*Pygoscelis adeliae*), Gentoo (*Pygoscelis papua*), and Chinstrap (*Pygoscelis antarcticus*). They are all helpful for exploring niche partitioning and morphological differentiation. This report is focused on two questions: Which morphological traits best distinguish Adelie, Chinstrap, and Gentoo penguins, and whether trait differences among species that coexist on the same island support niche partitioning. I explored patterns among penguins on islands and compared flipper length and body mass as body size, and bill length and bill depth as bill morphometrics in order to solve these two problems. These results indicate that, despite the fact that both Adelie and Chinstrap are found on Dream Island, body size is more beneficial for differentiating Gentoo penguins, while bill morphology is more useful for differentiating Adelie and Chinstrap. The paper indicates that feeding-related morphology, as opposed to overall body size alone, can offer more biological evidence for species differentiation and potentially niche partitioning.

Introduction

The Palmer Penguins dataset is an opportunity to research how morphology varies among closely related penguin species living in the Palmer Archipelago. Adelie, Gentoo, and Chinstrap penguins were recorded in the dataset. Although having diverse

body sizes, they shared a comparable regional environment, which might mean that they have distinct responsibilities in ecology. These differences are crucial because they might demonstrate how different penguins have adapted to their environment and decrease direct competition.

In biology, there is a concept called competitive exclusion, which means that two species competing for identical limiting resources cannot coexist indefinitely; the superior competitor will drive the other to extinction or force its adaptation. One of the possible results to explain this phenomenon is called niche partitioning. It is the strategy used by competitor species to reduce competition and increase coexistence by using the environment differently, such as by taking advantage of different resources or habitats. Features related to feeding morphology and body size in penguins may be valuable in determining whether biological differences between species are consistent with niche partitioning.

This report connects bill morphometrics with body-size variables using exploratory data analysis. Body size is compared by flipper length and body mass, and feeding-related morphology is compared by bill length and bill depth. This comparison facilitates identifying which characteristics offer more solid evidence of species differentiation and potentially niche partitioning.

Method

To solve the species identification problem, I determined the species names, bill length, bill depth, wing length, and body size. Then I divided these traits into two groups: body size (represented by wing length and body mass) and feeding-related morphology (represented by bill length and depth). I analyzed the morphometric index of species coexisting on the same island and used the island names to test for niche partitioning.

The analytical process uses R and the tidyverse suite for data manipulation, and ggplot2 for visualization. And the Palmer Penguins dataset exports current data as Comma-separated values (CSV) files. The data contains penguins from the Palmer Archipelago with 344 records with different variables: "Species", "Island", "Bill Length", "Bill Depth", "Flipper Length", "Body Mass", "Sex", and "Year". Although sex and year are parts of the dataset, neither of them will be a variable to study; Because other variables (species, island, bill length, bill depth, flipper length, and body mass) are more concentrated on the biological questions about species differentiation and niche partitioning.

To improve the accuracy of the exploratory computing process, here are the following steps:

1. Deal with Missing data: Almost all of the data was recorded in the original dataset. But unfortunately, some variables are still missing. For example, only the species and the year that were reported are shown in the fifth line; all other information is missing. The missing values for these essential variables were

removed before the data was shown. In order to reduce bias, the data is cleaned to make sure that all plots can compare the penguins with full morphological information. After removing rows with missing values in the selected morphological variables, 342 penguins remained for analysis.

	A	B	C	D	E	F	G	H
1	species	island	bill_length_mm	bill_depth_mm	flipper_length_m	body_mass_g	sex	year
2	Adelie	Torgersen	39.1	18.7	181	3750	male	2007
3	Adelie	Torgersen	39.5	17.4	186	3800	female	2007
4	Adelie	Torgersen	40.3	18	195	3250	female	2007
5	Adelie	Torgersen	NA	NA	NA	NA	NA	2007

2. Variable Chosen: The variables chosen were based on their biological meaning.

The island was used for the biological coexistence scenario, and species was used to establish the biological group for all penguins. Overall body size was determined by flipper length and body mass, and feeding-related morphometric characteristics were determined by bill length and bill depth. This comparison facilitates in determining whether bill morphology or overall body size provides greater proof for species difference.

3. Visual Exploration: Scatter plots were used to compare the relationships between species' traits. To measure the level of competitive exclusion between species, we transformed continuous morphological measurements into spatial clustering datasets. To determine whether body-size traits differentiate the three species, the first plot of flipper length and body mass was created.

4. Distribution analysis: Bill length distributions between species were compared using empirical cumulative distribution function (ECDF) plots. The next step determines whether the separation shown in scatter plots is present throughout the whole distribution compared to simply in a small number of individual observations.

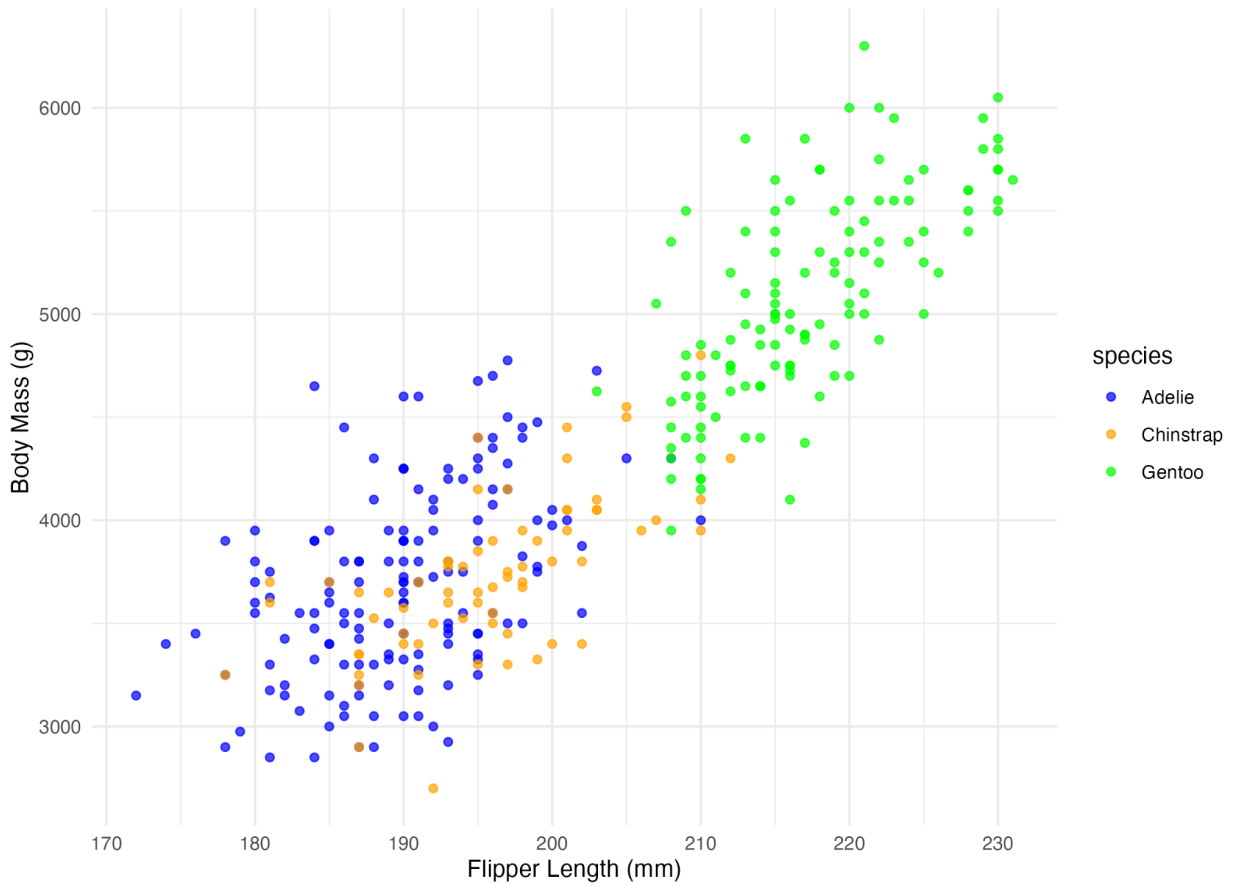
Result

1. Baseline Comparison: Flipper Length and Body Mass

To find out whether or not overall body size can differentiate the Adelie, Chinstrap, and Gentoo penguins, Figure 1 compares flipper length and body mass. With their larger bodies and longer flippers, Gentoo penguins form a separate cluster. Adelie and Chinstrap penguins, on the other hand, overlap in this body-size range.

This pattern demonstrates that body mass and flipper length are helpful in differentiating Gentoo penguins from the other two species, but they are insufficient in differentiating Adelie and Chinstrap penguins. As a result, although overall body size provides some indication of species differentiation, it is not enough to completely compensate for the variations across the three species.

Figure 1: Flipper Length vs Body Mass



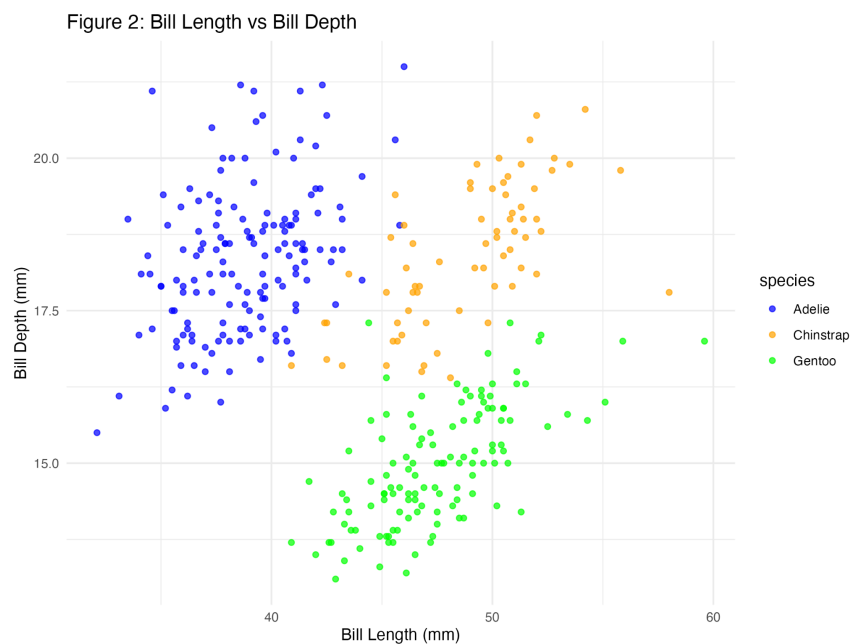
2. Primary Findings: Bill Morphometrics

Figure 2 compares bill length and bill depth to determine whether bill morphology provides better species separation than body-size characteristics. The three species demonstrate different patterns in bill morphospace compared to Figure 1. Gentoo penguins have longer but shallower bills, Adelie penguins have shorter and deeper bills, and Chinstrap penguins have longer and moderately deep bills.

These findings indicate that significant biological information for differentiating the three species can be found in bill length and bill depth. This division indicates that the

species may differ not only in appearance but also in characteristics connected to resource use because bill shape is related to feeding. Therefore, compared to total body size alone, bill morphometrics provide more evidence of species differentiation.

Each species' different eating habits provide strong evidence for this difference. Gentoo penguins have a broader diet that includes fish and squid, which explains their longer and shallower bill morphology, even if they all live in the same geographical habitat. On the other hand, through "prey-size selection" of Antarctic krill, Adelie and Chinstrap penguins decrease direct competition: Adelie usually prefer smaller, juvenile krill, while Chinstraps prefer larger, mature individuals. This provides evidence consistent with niche partitioning based on resource specialization and explains why their bill forms keep different even when they coexist on Dream Island.

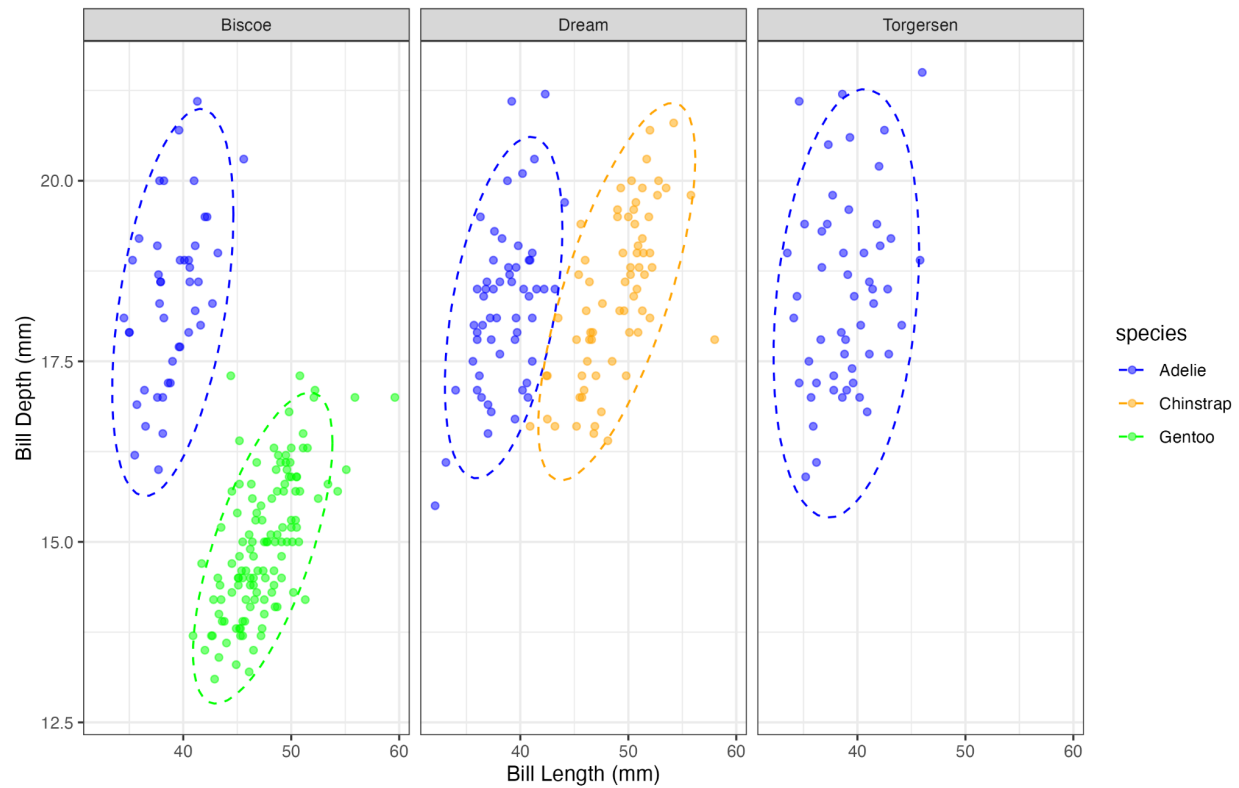


3. Evidence of Sympatric Adaptation by Island

Figure 3 focuses on bill length and bill depth across islands to figure out whether species that coexist in the same area continue to be physically distinct. Because of the presence of both Adelie and Chinstrap penguins on Dream Island, this comparison is very significant. These two species inhabit separate areas of bill-length and bill-depth space, although existing on the same island. Dream Island is especially useful for comparing coexisting species because the cleaned dataset has 68 Chinstrap penguins and 56 Adelie penguins.

The distribution of ecological niches follows the pattern even at the island level. If coexisting species used the same natural resources, it is expected that their feeding habits would overlap more. The Adelie and chinstrap penguins on Dream Island still have different beak shapes. Although the dataset does not specifically compare these species' diets, this morphological difference suggests the possibility that they use different ecological strategies to reduce direct competition.

Figure 3: Morphological Clusters by Species and Island

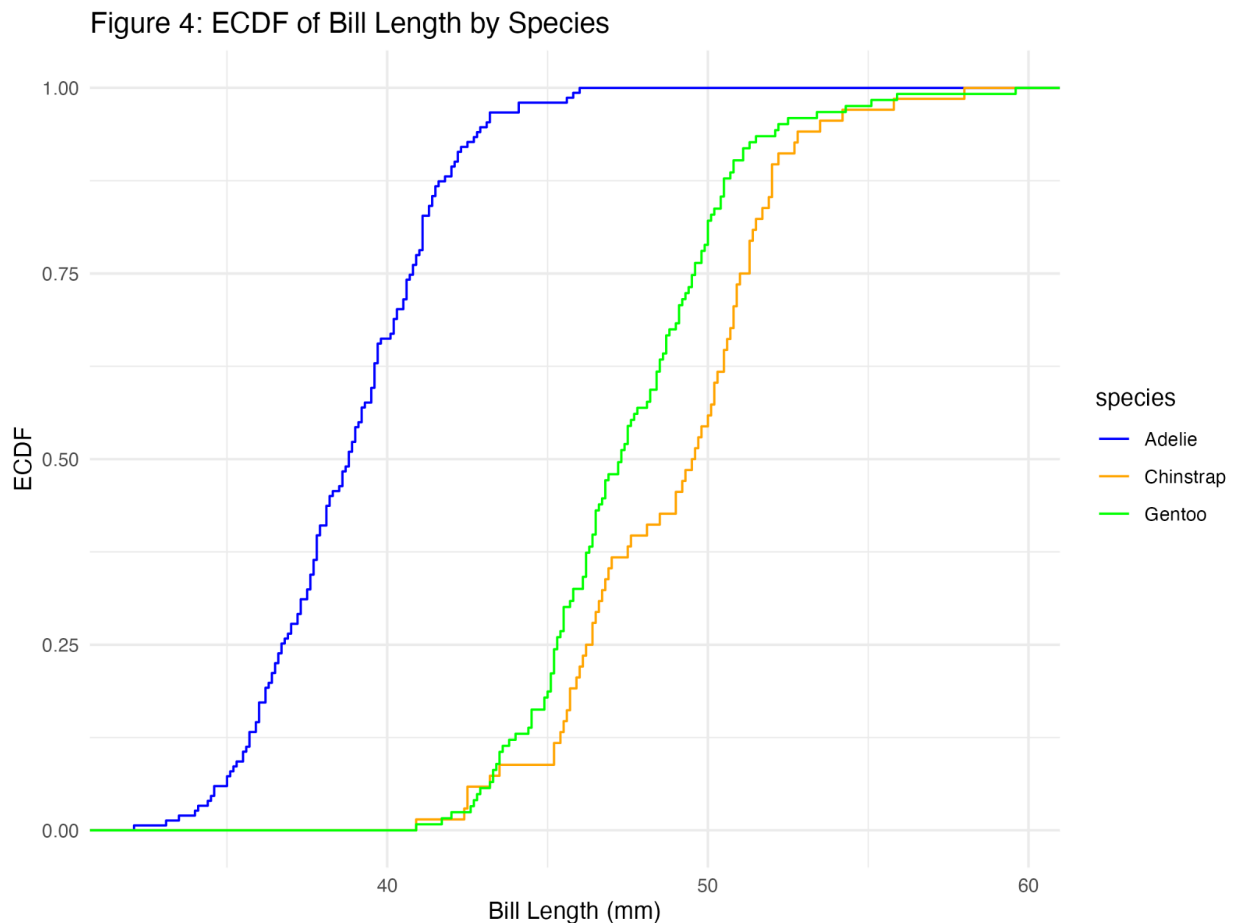


4. Distribution analysis: ECDF of bill lengths

Figure 4 compares the distribution of bill length in species using empirical cumulative distribution function plots. The ECDF curves are moved in relation to each other, showing that species variations in bill length are not restricted to a few extreme observations but rather exist throughout a large portion of the distribution. Chinstrap and Gentoo penguins have greater bill lengths than Adelie penguins.

The patterns shown in the scatter plots are supported by this distribution-level

evidence. Bill length reflects systematic distinction across species in contrast to individual differences. This indicates that bill morphology is one of the most effective sets of characteristics for determining penguin species differentiation, as well as bill depth.



Overall, the results show that bill morphometrics clearly distinguish the three species, especially Adelie and Chinstrap, while the size characteristics distinguish the Gentoo penguin. The continued separation of Adelie and Chinstrap on Dream Island provides biological evidence supporting niche partitioning.

Conclusion

This report indicates that different morphological factors provide different levels of biological information on the differentiation of penguin species. Because Gentoo penguins are typically larger and have longer flippers, body mass and flipper length can be used to identify them. However, Adelie and Chinstrap penguins cannot be differentiated from one another by these physical characteristics. On the other hand, bill depth and length make it easier to distinguish between the three species, especially between Adelie and Chinstrap.

The concept that bill morphology has biological significance is further supported by the island-level research. Adelie and Chinstrap penguins live together on Dream Island, however they are still separated by bill length and bill depth. Because coexisting species exhibit variations in feeding-related morphology, this pattern is associated with niche partitioning. This result should be considered as evidence compatible with niche partitioning rather than as direct evidence of different food sources because the dataset does not assess diet.

This analysis demonstrated that, depending on which features are analyzed, the same dataset could result in different biological explanations. Gentoo penguins were mostly differentiated from the other two species by body-size traits, while all three species were more clearly distinguished by bill length and depth. However, rather than being direct evidence of resource partitioning, the bill differences should be considered

to be evidence consistent with potential niche partitioning because the dataset does not include direct evaluations of food or foraging behaviors.

Reference:

Here are some references that helped me understand and use in my report:

(2026, April 7). *Understanding Scarcity in Economics: Causes and Effects*.

Investopedia. Retrieved April 7, 2026, from

<https://www.investopedia.com/terms/s/scarcity.asp>

(n.d.). 6.2.1: *Competitive exclusion principle*. LibreTexts Biology.

[https://bio.libretexts.org/Sandboxes/Team%3A_Ecology_Wildlife_and_Conservation_\(EVC\)/09%3A_Species_Interactions_in_Communities/9.02%3A_Competitive_Interactions/9.2.01%3A_Competitive_exclusion_principle](https://bio.libretexts.org/Sandboxes/Team%3A_Ecology_Wildlife_and_Conservation_(EVC)/09%3A_Species_Interactions_in_Communities/9.02%3A_Competitive_Interactions/9.2.01%3A_Competitive_exclusion_principle)

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(n.d.). *Niche Partitioning*. Earth and Planetary Sciences.

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Trivelpiece, W. Z., Trivelpiece, S. G., & Volkman, N. J. (1987). Ecological Segregation of Adelie, Gentoo, and Chinstrap Penguins at King George Island, Antarctica. *Ecology*, 68(2), 351–361. <https://doi.org/10.2307/1939266>

Class materials: Discussion 2,3 and Lecture slides

Appendix:

R code:

```
# STA 160 Mid-Quarter Project

# Topic: Palmer Penguin Niche Partitioning: A Data-Driven Analysis of Morphological
Differentiation

# Jiawei Zhu

# 1. Load Necessary Libraries

library(palmerpenguins)

library(tidyverse)

# Save the original dataset to a CSV file, that to show there are some NA values in
the original dataset, and to make it easier for others to access the data without
```

```

needing to load the palmerpenguins package.

write.csv(penguins, "penguins.csv", row.names = FALSE)

# For Data Cleaning

# Remove rows with NA values in the relevant columns to ensure data integrity

new_penguins <- penguins %>%

  filter(!is.na(bill_length_mm) & !is.na(bill_depth_mm) &

         !is.na(flipper_length_mm) & !is.na(body_mass_g))

nrow(new_penguins)

# Visualizations Data Cleaning

# First Graph: Flipper Length vs Body Mass (Low Explanatory Power due to overlap)

new_body <- ggplot(new_penguins, aes(x = flipper_length_mm, y = body_mass_g, color =
species)) +

  geom_point(alpha = 0.7) +

  labs(title = "Figure 1: Flipper Length vs Body Mass",

        x = "Flipper Length (mm)",

        y = "Body Mass (g)") +

  theme_minimal() +

  scale_color_manual(values = c("Adelie" = "blue", "Chinstrap" = "orange", "Gentoo" =
"green"))

```

```

# Second Graph: Bill Morphometrics (High Explanatory Power - Clear Clusters)

new_bill <- ggplot(new_penguins, aes(x = bill_length_mm, y = bill_depth_mm, color =
species)) +

  geom_point(alpha = 0.7) +

  labs(title = "Figure 2: Bill Length vs Bill Depth",

        x = "Bill Length (mm)",

        y = "Bill Depth (mm)") +

  theme_minimal() +

  scale_color_manual(values = c("Adelie" = "blue", "Chinstrap" = "orange", "Gentoo" =
"green"))

# Third Graph: Advanced Morphological Clusters by Island (Sympatric Evidence)

# Adding 95% confidence ellipses and faceting by island

new_advanced <- ggplot(new_penguins, aes(x = bill_length_mm, y = bill_depth_mm, color
= species)) +

  geom_point(alpha = 0.5) +

  stat_ellipse(level = 0.95, linetype = 2) +

  facet_wrap(~island) +

  theme_bw() +

  labs(title = "Figure 3: Morphological Clusters by Species and Island",

        x = "Bill Length (mm)",

        y = "Bill Depth (mm)") +

  scale_color_manual(values = c("Adelie" = "blue", "Chinstrap" = "orange", "Gentoo" =
"green"))

# Count of Each Species on Dream Island

```



```

dreams_penguins <- new_penguins %>%

  filter(island == "Dream") %>%

  count(species)

dreams_penguins

# Fourth Graph: ECDF of Bill Length by Species (Additional Insight into Morphological
Differentiation)w

new_ecdf <- ggplot(new_penguins, aes(x = bill_length_mm, color = species)) +

  geom_step(stat = "ecdf") +

  labs(title = "Figure 4: ECDF of Bill Length by Species",

        x = "Bill Length (mm)",

        y = "ECDF") +

  theme_minimal() +

  scale_color_manual(values = c("Adelie" = "blue", "Chinstrap" = "orange", "Gentoo" =
"green"))

# For easily to check the graphs

ggsave(plot = new_body, filename = "Figure_1.png", width = 8, height = 6, dpi = 300)

ggsave(plot = new_bill, filename = "Figure_2.png", width = 8, height = 6, dpi = 300)

ggsave(plot = new_advanced, filename = "Figure_3.png", width = 9, height = 6, dpi =
300)

ggsave(plot = new_ecdf, filename = "Figure_4.png", width = 8, height = 6, dpi = 300)

```